

PINEGA, Russian Federation

WMO: 225630

Lat: 64.700N

Lon: 43.383E

Elev: 31

StdP: 100.96

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.5	-30.7	-37.4	0.1	-34.6	-33.4	0.2	-30.6	7.1	-6.9	6.3	-7.4	0.9	330	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.3	18.7	25.1	17.7	23.0	16.6	20.1	25.3	18.8	23.4	17.6	21.8	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.3	22.3	17.1	12.2	21.0	15.7	11.2	19.6	57.7	25.3	53.6	23.5	49.7	21.7	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.9	5.2	-36.2	29.8	3.5	2.1	-38.8	31.3	-40.8	32.5	-42.8	33.7	-45.4	35.2		
(5)			-36.3	21.6	3.5	1.9	-38.9	23.0	-40.9	24.1	-42.9	25.1	-45.4	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-13.5	-12.5	-6.9	0.4	6.6	12.0	15.4	13.0	8.1	1.8	-5.0	-9.5	(6)
(7)		DBStd	11.51	8.82	8.55	6.12	4.77	5.27	4.71	4.35	4.02	3.55	4.05	6.60	8.30	(7)
(8)		HDD10.0	3742	727	629	524	289	133	30	6	14	78	256	450	605	(8)
(9)		HDD18.3	6395	986	863	782	538	363	197	111	171	308	513	700	863	(9)
(10)		CDD10.0	425	0	0	0	1	29	91	174	108	21	1	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	1	8	21	7	0	0	0	0	(11)
(12)		CDH23.3	394	0	0	0	0	14	97	214	68	1	0	0	0	(12)
(13)		CDH26.7	76	0	0	0	0	1	22	40	13	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.4	2.5	2.6	2.7	2.8	2.5	2.4	2.3	2.3	2.7	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	557	30	25	29	34	38	68	64	70	59	56	48	39	(15)
(16)		PrecMax	743	71	61	76	70	75	174	125	147	100	98	107	66	(16)
(17)		PrecMin	424	7	3	6	7	5	1	24	16	11	7	17	12	(17)
(18)		PrecStd	85	13	13	15	17	17	38	29	31	23	22	20	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.5	7.2	18.2	25.2	29.2	29.9	28.4	20.8	12.6	6.4	2.8	(19)
(20)			MCWB	0.5	1.6	3.1	10.1	15.5	19.1	20.2	20.0	15.5	10.3	5.7	2.2	(20)
(21)		2%	DB	0.3	1.0	3.8	12.2	21.6	26.2	27.7	25.1	17.8	10.0	4.6	1.5	(21)
(22)			MCWB	-0.1	0.4	1.1	6.8	13.2	17.5	18.9	18.5	14.0	8.6	4.2	0.9	(22)
(23)		5%	DB	-0.8	0.0	2.4	9.1	18.9	23.4	25.9	22.2	15.7	8.4	3.1	0.9	(23)
(24)			MCWB	-1.3	-0.6	0.5	4.8	11.9	15.5	18.6	17.0	12.8	7.4	2.7	0.4	(24)
(25)		10%	DB	-2.4	-1.6	1.2	7.0	16.3	20.9	23.9	20.0	13.8	7.0	1.7	0.1	(25)
(26)			MCWB	-2.9	-2.2	-0.1	3.6	10.5	14.5	17.6	16.3	11.9	6.2	1.2	-0.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.6	3.4	11.0	17.0	20.1	22.5	21.3	16.3	10.9	6.0	2.2	(27)
(28)			MCDWB	1.0	2.2	6.3	16.7	23.6	27.3	27.6	26.0	19.3	11.9	6.3	2.5	(28)
(29)		2%	WB	-0.1	0.5	1.8	7.4	14.5	18.5	20.6	19.4	14.6	8.9	4.2	1.0	(29)
(30)			MCDWB	0.3	0.9	3.2	11.5	19.5	24.3	25.3	23.7	17.0	9.9	4.5	1.3	(30)
(31)		5%	WB	-1.3	-0.6	0.9	5.3	12.6	17.0	19.5	17.8	13.3	7.5	2.7	0.4	(31)
(32)			MCDWB	-0.9	-0.1	1.9	8.2	17.9	21.6	24.1	21.2	15.2	8.3	3.1	0.8	(32)
(33)		10%	WB	-2.9	-2.1	0.0	3.9	10.9	15.2	18.2	16.6	12.0	6.3	1.3	-0.4	(33)
(34)			MCDWB	-2.4	-1.5	1.2	6.6	15.8	19.5	22.5	19.5	13.7	7.0	1.7	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	7.4	9.3	8.9	9.7	10.0	9.9	8.6	7.1	4.2	4.3	5.4	(35)
(36)			MCDBR	5.5	5.3	8.1	12.1	15.0	14.1	13.6	12.2	10.8	5.6	3.8	3.7	(36)
(37)			MCWBR	5.4	5.0	6.1	7.5	8.0	6.5	6.4	6.6	6.9	4.5	3.8	3.7	(37)
(38)		5% WB	MCDBR	5.4	5.4	6.1	10.9	13.5	12.1	11.7	10.8	9.3	5.3	3.8	3.5	(38)
(39)			MCWBR	5.3	5.2	4.8	7.3	7.7	6.3	6.4	6.6	6.3	4.6	3.8	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.220	0.281	0.310	0.345	0.350	0.347	0.386	0.360	0.331	0.305	0.356	0.217	(40)	
(41)		taud	1.935	2.061	2.112	2.157	2.422	2.484	2.383	2.435	2.503	2.347	2.576	1.947	(41)	
(42)		Ebn at Noon	290	591	752	817	859	870	816	805	756	589	282	83	(42)	
(43)		Edh at Noon	25	63	94	115	98	94	102	88	67	49	23	9	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.13	0.66	2.00	3.69	4.71	5.25	5.03	3.43	1.88	0.67	0.16	0.04	(44)	
(45)		RadStd	0.02	0.12	0.26	0.44	0.41	0.49	0.53	0.36	0.19	0.11	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page